



AI ENGINEERING GUIDE

Token Optimization Techniques

**Reduce Cost. Improve Latency.
Scale AI Systems.**

A practical guide for AI Engineers building
production AI applications.

Swipe to learn all 10 techniques →



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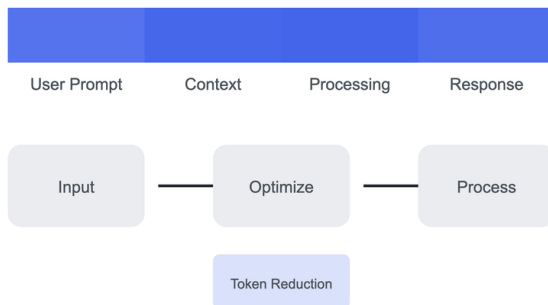
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The Production AI Engineer's **Token Optimization Toolkit**

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| 01 Prompt Compression
Remove unnecessary verbosity | 02 Context Filtering
Send only relevant info |
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JSON instead of long text |
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Prevent context waste |

Token Budget Allocation



Prompt Compression

Many prompts contain unnecessary words that inflate token count without improving output quality.

✖ BAD

"Please carefully analyze the following information and provide a detailed response."

✔ BETTER

"Analyze and summarize."

💡 OPTIMIZATION IDEAS

- Remove repeated instructions
- Shorten system prompts
- Eliminate unnecessary examples
- Use reusable prompt templates



20–50% token reduction in prompts with tighter instructions.



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Context Filtering

Most systems send too much information. Only a fraction is actually useful.

10

RETRIEVED CHUNKS



3

ACTUALLY USEFUL

▼ FILTER CONTEXT USING

- ✓ Similarity score thresholds
- ✓ Metadata filters
- ✓ Access permissions
- ✓ Re-ranking models

↓ Less context. Better answers. Lower cost.



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TECHNIQUE 03

Dynamic Context Injection

Many systems always send everything regardless of what the user actually needs.

✖ ALWAYS INJECTED (PROBLEM)

- ✖ User Profile
- ✖ Policies
- ✖ Documentation
- ✖ Billing Docs
- ✖ Product Data
- ✖ Chat History

✔ INJECT ONLY WHAT FITS THE QUERY

User asks: "Reset my password."

✔ Auth docs

✖ Billing docs

✖ Pricing docs



Significant token savings on every single request.



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Semantic Caching

The same questions appear repeatedly in different phrasings. Most systems process each one from scratch.

REPEATED QUERIES (PROBLEM)

- 🗨 "Reset my password"
- 🗨 "Change password"
- 🗨 "Forgot password"

CACHE SOLUTION

Query Embedding



Cache Store



Reuse Response

BENEFITS

- ✓ Faster responses on repeated queries
- ✓ Lower API cost per request
- ✓ Reduced model usage overall



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Conversation Summarization

Long conversations consume large context windows and drive up cost with every turn.

⚠ BEFORE

100

Messages

~8,000

Tokens per call

✅ AFTER

Summary

~500

Recent Messages

~300

Total tokens

⚙ HOW IT WORKS

- Summarize older turns into a compact block
- Keep only the most recent N messages verbatim
- Prepend summary at the start of each new call



Keep context quality while dramatically reducing token usage.



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Context Minification

Retrieved documents are often much larger than necessary. Most of the content is irrelevant to the query.

🔍 EXTRACT ONLY WHAT MATTERS

- ✓ Relevant paragraphs
- ✓ Key facts and figures
- ✓ Important sections

📄 BEFORE

5,000

token document



📄 AFTER

700

token summary



Smaller prompts. Faster generation. Lower cost per call.



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RAG Optimization

More chunks does not mean better answers. Poor retrieval inflates tokens and degrades quality.

❌ BAD RAG

20

chunks retrieved

8,000

tokens

✅ GOOD RAG

3

highly relevant chunks

1,200

tokens

⚙️ OPTIMIZATION TECHNIQUES

- ✓ Better chunking strategies
- ✓ Re-ranking retrieved results
- ✓ Hybrid search (dense + sparse)
- ✓ Metadata filters before retrieval



Higher retrieval quality with significantly fewer tokens.



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Structured Outputs

LLMs often generate unnecessary explanations and preamble. Force structured output to eliminate token waste.

→ Instead, generate:

```
{  
  "priority": "high",  
  "status": "approved",  
  "risk": "low"  
}
```

✓ BENEFITS

- ✓ Smaller output responses
- ✓ Easier downstream automation
- ✓ Better reliability and parsing

↓ Less output token consumption on every generation call.



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Model Routing

Not every task needs the largest model. Routing to smaller models saves significant cost.



Spam Detection



Small Model



Complex Planning



Large Model

ROUTING FLOW

User Query



Classifier



Best Model



Lower cost while maintaining the quality users expect.



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Token-Aware Agent Design

Agents often waste tokens through poor memory and planning patterns.

❗ COMMON ISSUES

- ✗ Re-reading documents each step
- ✗ Repeating tool outputs in prompt
- ✗ Sending full memory every step
- ✗ Duplicate reasoning loops

✅ SOLUTIONS

- ✓ Memory compression
- ✓ Selective retrieval
- ✓ Tool output summarization
- ✓ Smart step planning



More efficient agent workflows with lower cost per task.



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KEY TAKEAWAY

The Best Token Is The One You Never Send

HIGHEST ROI TECHNIQUES

#1  Context Filtering

#2  Dynamic Context Injection

#3  Conversation Summarization

#4  Semantic Caching

#5  RAG Optimization

 Production AI systems commonly reduce token usage by **30–70%** using these techniques.



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